

Point

From Number Theory to Algebraic Geometry and Logic

Detailed Syllabus — Version 2.1 (Merged Maximal Edition)

64 weekly meetings (2–2.5 h): P1–P3 and S1–S61

Every session: union of all content, proofs, course-role, thread, and references

Abstract

This is the *merged maximal* edition of the syllabus: for each of the 64 meetings it records the union of all teaching content, the theorems proved in class, the session’s role in the course (*Depends on / Feeds / Arc*), its incarnation of the unifying concept of a *point*, and consolidated references. The course follows the point from primes, through valuations, ideals, T -valued points, places, geometric morphisms, and models, to Deligne’s theorem that every coherent topos has enough points. All main theorems are proved; all prerequisites are built inside the course, just in time.

Anatomy of a session entry

Each entry has five labelled parts: **Content to cover** (the complete teaching list); **Main results proved**; **Role in the course** (*Depends on / Feeds / Arc*); **Unifying thread**; **References**.

Contents

Structure at a Glance	4
1 Prelude — Essential Prerequisites	5
1.1 P1. Group Theory: Part I	5
1.2 P2. Topology and Metric Spaces	5
1.3 P3. Algebraic Topology Crash Course	6
2 Act 0 — Foundations	6
2.1 1. Rings, Ideals, Quotients, Prime & Maximal Ideals	6
2.2 2. Modules, Tensor Products, and Exact Sequences	6
2.3 3. Fields, Extensions, and Algebraic Closure	7
2.4 4. Category Theory Essentials	7
3 Act I — The Point as a Prime Number	7
3.1 5. Quadratic Residues and the Legendre Symbol	7
3.2 6. Quadratic Reciprocity: Statement and First Proof	8
3.3 7. Quadratic Reciprocity: Geometric Proofs & Generalizations	8
3.4 8. Reciprocity in the Language of Galois Theory	8
3.5 9. Cyclotomic Extensions and Frobenius Elements	9
3.6 10. Summary: From Reciprocity to Galois Theory	9

4	Act II — Galois Theory and Solvability	9
4.1	11. Solvable Groups, S_n , and A_n	9
4.2	12. Galois Extensions, Finite Fields, and Linear Independence	10
4.3	13. The Fundamental Theorem of Galois Theory	10
4.4	14. Solvable Groups and Solvability by Radicals	10
4.5	15. The Insolubility of the Quintic	10
4.6	16. The Absolute Galois Group	11
5	Act III — Valuations and p-adic Points	11
5.1	17. Absolute Values, Valuations, and Ostrowski's Theorem	11
5.2	18. Construction of $\mathbb{Q}_p$ and $\mathbb{Z}_p$	11
5.3	19. Hensel's Lemma and p -adic Topology	12
5.4	20. Extension of Valuations: Ramification and Residue Degree	12
5.5	21. Function Fields and the Geometric Analogy	12
5.6	22. Adeles, Ideles, and Adelic Compactness	13
6	Act IV — Algebraic Geometry	13
6.1	23. Commutative Algebra I	13
6.2	24. Dedekind Domains and Ideal Class Groups	13
6.3	25. Affine Varieties and Hilbert's Nullstellensatz	14
6.4	26. Presheaves and Sheaves	14
6.5	27. Sheafification, Categories of Sheaves, and Étale Spaces	14
6.6	28. Schemes: $\text{Spec}(R)$, the Structure Sheaf, Closed & Generic Points	15
6.7	29. The Functor of Points and Yoneda's Perspective	15
6.8	30. Commutative Algebra II: Morphisms, Fiber Products, Étale Maps	15
6.9	31. Projective Varieties and Points at Infinity	16
6.10	32. Divisors, the Product Formula, and Riemann–Roch (Algebraic)	16
6.11	33. The Valuative Criterion of Properness	16
6.12	34. Free Groups and Profinite Groups	17
6.13	35. Étale Fundamental Group and Grothendieck's Galois Theory	17
6.14	36. Summary: The Bridge Between Number Theory and Geometry	17
7	Act V — Curves over $\mathbb{C}$ and Elliptic Curves	18
7.1	37. Complex Analysis I: Holomorphic Functions & Cauchy Theory	18
7.2	38. Complex Analysis II: Riemann Surfaces	18
7.3	39. Elliptic Curves: Geometry and Group Law	18
7.4	40. Elliptic Curves: Arithmetic and Modular Forms	19
7.5	41. Points as Objects: Moduli Spaces and Rational Points	19
8	Act VI — Class Field Theory	19
8.1	42. Homological Algebra I: Complexes and the Snake Lemma	19
8.2	43. Homological Algebra II: Resolutions and Derived Functors	20
8.3	44. Group Cohomology and Tate Cohomology	20
8.4	45. Galois Cohomology and the Fundamental Exact Sequence	20
8.5	46. The Artin Map	20
8.6	47. Artin Reciprocity	21
8.7	48. The Existence Theorem	21
8.8	49. The Idele Formulation and Global CFT	21
8.9	50. Local Class Field Theory: Lubin–Tate	22
9	Interlude — Spaces without Points	22
9.1	51. Gelfand–Naimark and the Non-commutative Glimpse	22

10 Act VII — Topos Theory and Logic	22
10.1 52. Sites and Grothendieck Topologies	22
10.2 53. Sheaves on Sites and Topoi	23
10.3 54. Geometric Morphisms and Points of a Topos	23
10.4 55. Enough Points	23
10.5 56. Classifying Topoi and Geometric Theories	24
10.6 57. Geometric Model Theory	24
10.7 58. Deligne’s Theorem: Statement and Proof	24
10.8 59. Deligne’s Theorem in Algebraic Geometry	25
11 Postlude — Synthesis and the Big Picture	25
11.1 60. The Point: One Concept, Many Worlds	25
11.2 61. Points in Mechanics, Quantum Physics, and Langlands	25
Consolidated References	26

Structure at a Glance

Part	Title	Sessions	Theme
Prelude	Prerequisites	P1–P3	Groups; topology & metric; algebraic topology
Act 0	Foundations	1–4	Rings/spectra; modules/tensor; fields; categories
Act I	Point as a Prime	5–10	Quadratic reciprocity & Frobenius
Act II	Galois Theory	11–16	Solvability, FTGT, absolute Galois group
Act III	Valuations & p -adics	17–22	Ostrowski, $\mathbb{Q}_p$, adèles & compactness
Act IV	Algebraic Geometry	23–36	Comm. algebra, sheaves, schemes, functor of points, étale π_1
Act V	Curves over $\mathbb{C}$	37–41	Complex analysis, Riemann surfaces, elliptic curves, moduli
Act VI	Class Field Theory	42–50	Homological engine, cohomology, reciprocity, Lubin–Tate
Interlude	Spaces without Points	51	Gelfand–Naimark, non-commutative glimpse
Act VII	Topos & Logic	52–59	Sites, topoi, points, classifying topoi, Deligne
Postlude	Synthesis	60–61	One concept many worlds; physics & Langlands

Prerequisite map (introduced $\rightarrow$ first used)

Tool	Introduced	First used
Zorn's lemma	S1	S3, S58
π_1 & covering spaces	P3	S35
Modules, tensor, Nakayama	S2	S23, S30, S42
Yoneda lemma	S29	S41, S54
Cyclicity of $(\mathbb{Z}/p)^\times$	S5	S5
Dedekind's lemma	S12	S13, S14, S20, S45
Noetherian, integral closure	S23	S24, S25, S28
Sheaf theory	S26–S27	S28, S53
Fiber products, étale	S30	S33, S35, S39
Restricted product, adelic compactness	S22	S24, S49
Complex analysis	S37–S38	S39, S40
Homological algebra	S42–S43	S44, S45
Formal group laws	S50	S50

1 Prelude — Essential Prerequisites

1.1 Session P1 — Group Theory: Part I

Content to cover.

- Groups and examples: $\mathbb{Z}$, $\mathbb{Q}^*$, S_n , $\mathrm{GL}_n(k)$, $\mathrm{SO}_n(\mathbb{R})$, circle group; abelian vs non-abelian; orders.
- Subgroups; subgroup criterion; generated & cyclic subgroups; order of an element; Lagrange + corollaries (order divides; $g^n = e$; prime order $\Rightarrow$ cyclic).
- Homomorphisms; kernel/image; injectivity via trivial kernel; isomorphisms; inner automorphisms & center.
- Normal subgroups & criteria; quotient groups (well-definedness); natural projection; First/Second/Third Isomorphism Theorems.
- Three worked First-Iso examples: $\det$, sign , $\mathbb{Z} \rightarrow \mathbb{Z}/n$.
- Group actions; orbits, stabilizers; Orbit–Stabilizer; conjugation action; class equation (mention); Cayley (mention).

Main results proved. Subgroup criterion; Lagrange; First Isomorphism; Orbit–Stabilizer.

Role in the course. *Depends on* none (entry). *Feeds* Act II, S24, S49, P3/S35, Act VII. *Arc*: quotient = identify what looks the same from a viewpoint — the pattern later instantiated by norm groups and $G_K^{\mathrm{ab}} \cong C_K$.

Unifying thread. *A quotient group is a space of viewpoints collapsed to points.*

References. Artin ch.2; Herstein ch.2; Rotman ch.1–4.

1.2 Session P2 — Topology and Metric Spaces

Content to cover.

- Topology axioms; open/closed; bases; subspace & product topologies; infinite products vs box (remark previewing restricted product, S22).
- Continuity (open-set); homeomorphism; connectedness & components; intervals connected (proof); compactness; $[0, 1]$ compact (bisection proof); Heine–Borel (statement); Hausdorff.
- Metric spaces; balls; metric topology; convergence; Cauchy; completeness; examples $\mathbb{R}$ complete, $\mathbb{Q}$ not, $C[0, 1]$ sup complete.
- Completion (Cauchy mod null): existence & uniqueness; $\mathbb{Q} \rightarrow \mathbb{R}$.
- Preview: p -adic metric & $\mathbb{Q} \rightarrow \mathbb{Q}_p$ (S17–S18); Ostrowski preview (S17).

Main results proved. Bases generate topology; balls a basis; intervals connected; $[0, 1]$ compact; compact $\subseteq$ Hausdorff closed; unique limits; ε – $\delta \iff$ open-set continuity; convergent $\Rightarrow$ Cauchy; completion existence/uniqueness.

Role in the course. *Depends on* none. *Feeds* S18, S19, S21, S22, S34, S38. *Arc*: supplies the two machines the course runs on — completion (adding missing points) and compactness (finiteness).

Unifying thread. *Completion adds the points Cauchy sequences deserved; compactness makes point-counts finite.*

References. Munkres ch.2–4; Sutherland; Rudin ch.2–4; Gouvêa ch.1–3.

1.3 Session P3 — Algebraic Topology Crash Course

Content to cover.

- Paths & homotopy of paths; homotopy equivalence; $\pi_1(X, x_0)$; computations for S^1 and the torus; simply-connected spaces.
- Covering spaces; lifting of paths & homotopies; deck transformations; universal cover; classification of coverings via π_1 -sets.
- The topological Galois correspondence: covers $\leftrightarrow$ subgroups / π_1 -sets (prototype of S35).
- Monodromy: lifting a loop permutes the fiber — a first “evaluation along a loop”.
- Scope note: no homology; just enough to make étale π_1 intuitive; remark on profinite completion & the $G_{\mathbb{Q}}$ analogy (S16).

Main results proved. $\pi_1(S^1) = \mathbb{Z}$; the covering-space correspondence (construction).

Role in the course. *Depends on* P2. *Feeds* S35, S59, S60. *Arc:* provides the exact categorical pattern (covers $\leftrightarrow$ π_1 -sets) that S35 reproduces for schemes and Act VII internalizes in topoi.

Unifying thread. *A covering is a space of points over points; monodromy is evaluation along loops.*

References. Hatcher ch.0–1 (selected); May ch.1–3 (selected); Munkres §51–54; Szamuel ch.2.

2 Act 0 — Foundations

2.1 Session 1 — Rings, Ideals, Quotients, Prime & Maximal Ideals

Content to cover.

- Zorn’s lemma; Krull’s theorem (existence of maximal ideals) [prereq for S3, S58].
- Rings, ideals, quotient rings; localization R_S , $R_{\mathfrak{p}}$; examples $\mathbb{Z}$, $k[x]$, $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-5}]$.
- Prime ideals ($R/\mathfrak{p}$ a domain); maximal ideals ($R/\mathfrak{m}$ a field); maximal $\Rightarrow$ prime (proofs); nilradical $= \bigcap$ primes (proof).
- $\text{Spec}(R)$; Zariski topology via $V(I)$; basic opens $D(f)$ form a basis; quasi-compactness (proof).
- Closed points = maximal; generic point = minimal prime; specialization; examples $\text{Spec } \mathbb{Z}$, $\text{Spec } \mathbb{C}[x]$, $\text{Spec } \mathbb{C}[x, y]$, $\text{Spec } \mathbb{Z}_{(p)}$.
- Functoriality $\text{Spec} : \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Top}$; remark: why Zariski is not Hausdorff (preview S28).

Main results proved. Maximal/prime characterizations; Krull; nilradical intersection; Zariski axioms; $D(f)$ basis; Spec quasi-compact.

Role in the course. *Depends on* none. *Feeds* S28, S46, S54; Zorn reused in S3, S58. *Arc:* the first incarnation of point.

Unifying thread. *Prime ideal = point; the generic point is “the whole space seen as one point”.*

References. Atiyah–Macdonald ch.1–4; Eisenbud ch.2; Vakil ch.3–4.

2.2 Session 2 — Modules, Tensor Products, and Exact Sequences

Content to cover.

- Modules, submodules, quotients; free & finitely generated; examples.
- Nakayama’s lemma (proof) + applications to local rings & generators.
- Tensor product $M \otimes_R N$; universal property; right-exactness (proof); base change; $M \otimes_R k(\mathfrak{p})$ as the fiber at $\mathfrak{p}$.

- Short & long exact sequences; split sequences; diagram-chasing warm-up (snake deferred to S42).
- Flatness: definition & first examples (free, localized); preview of its role in étale maps (serious treatment S30).

Main results proved. Nakayama; right-exactness of tensor; base-change identities.

Role in the course. *Depends on* S1. *Feeds* S23, S30, S42–S45. *Arc:* tensoring with a residue field = evaluating a family at a point.

Unifying thread. *The fiber $M \otimes_R k(\mathfrak{p})$ is the value of the module at the point $\mathfrak{p}$.*

References. Atiyah–Macdonald ch.2–3; Eisenbud ch.1,6; Lang ch.XVI.

2.3 Session 3 — Fields, Extensions, and Algebraic Closure

Content to cover.

- Fields; characteristic; finite fields $\mathbb{F}_p, \mathbb{F}_{p^n}$ (existence/uniqueness deferred to S12).
- Extensions L/K , degree, tower law; algebraic vs transcendental; minimal polynomial.
- Splitting fields (preview); algebraic closure $\bar{K}$: existence via Zorn (proof), uniqueness (mention); $\bar{\mathbb{Q}}$.
- Definition of $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ as the horizon object (developed S16).

Main results proved. Tower law; existence of algebraic closures.

Role in the course. *Depends on* S1 (Zorn). *Feeds* Act II, S16, S35. *Arc:* $\bar{\mathbb{Q}}$ is the space of all algebraic points over $\mathbb{Q}$.

Unifying thread. *Algebraic closure gathers all points of all extensions into one space.*

References. Lang ch.V; Artin, *Galois Theory* ch.1–3.

2.4 Session 4 — Category Theory Essentials

Content to cover.

- Categories, functors, natural transformations; examples **Set**, **Ring**, **Top**, $R\text{-Mod}$.
- Limits & colimits concretely: products, equalizers, inverse limits ($\mathbb{Z}_p$, profinite groups, $G_{\mathbb{Q}}$), directed colimits (stalks, localization).
- Adjunctions & universal properties (free/forgetful; preview $f^{-1} \dashv f_*$); Spec as $\mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Top}$.
- Yoneda lemma: statement & preview (full proof S29).
- Remark (one line): monads/comonads — sheafification (S27) is an idempotent monad; full monad theory not pursued.

Main results proved. Worked (co)limit computations; adjunction examples.

Role in the course. *Depends on* S1–S2. *Feeds* S16, S18, S26, S29, S34, S53–S54. *Arc:* an object is determined by its maps in/out — precursor of “point = morphism”.

Unifying thread. *Limits glue local data into global objects; colimits extract germs at points.*

References. Mac Lane ch.I–V; Leinster.

3 Act I — The Point as a Prime Number

3.1 Session 5 — Quadratic Residues and the Legendre Symbol

Content to cover.

- Congruences recap; quadratic residues; the Legendre symbol (a/p) .
- Cyclicity of $(\mathbb{Z}/p)^\times$ (primitive roots); Euler’s criterion (proof); multiplicativity.
- Supplementary laws $(-1/p)$, $(2/p)$ (proofs); extensive small-prime computations.
- The quadratic character as a homomorphism $(\mathbb{Z}/p)^\times \rightarrow \{\pm 1\}$ — the first “evaluation character”.

Main results proved. Primitive roots; Euler’s criterion; supplementary laws.

Role in the course. *Depends on P1. Feeds S6–S8, S46, S61. Arc:* the prime p acts as a place evaluating squareness.

Unifying thread. (a/p) is the value of a at the point p .

References. Ireland–Rosen ch.5; Hardy–Wright ch.VI.

3.2 Session 6 — Quadratic Reciprocity: Statement and First Proof

Content to cover.

- Precise statement of QR; Gauss’s Lemma (counting least residues).
- Complete proof of QR via Gauss’s Lemma; numerical examples.
- Reformulation preview: (p/q) as splitting of q in $\mathbb{Q}(\sqrt{p})$.

Theorem (Gauss, 1796). *For distinct odd primes p, q : $(p/q)(q/p) = (-1)^{(p-1)(q-1)/4}$.*

Main results proved. QR via Gauss’s Lemma.

Role in the course. *Depends on S5. Feeds S7–S8, S47. Arc:* the theorem whose generalization is the whole course.

Unifying thread. *Two primes evaluate each other; the answers agree up to a sign rule.*

References. Ireland–Rosen ch.5; Cox ch.1.

3.3 Session 7 — Quadratic Reciprocity: Geometric Proofs & Generalizations

Content to cover.

- Lattice-point (geometric) proof of QR (proof); Eisenstein’s trigonometric-sum proof (mention).
- Jacobi symbol; QR for Jacobi; computational applications.
- Remark: reciprocity as a point-counting statement (preview of L -functions S40 and Langlands S61).

Main results proved. Geometric proof of QR.

Role in the course. *Depends on S6. Feeds S8, S40, S61. Arc:* shows reciprocity has geometric content.

Unifying thread. *Counting points mod p packages into the reciprocity law.*

References. Ireland–Rosen ch.5; Lemmermeyer.

3.4 Session 8 — Reciprocity in the Language of Galois Theory

Content to cover.

- Pre-Galois toolkit (self-contained): $\text{Gal}(L/K) = \text{Aut}(L/K)$; concrete Galois groups of $\mathbb{Q}(\sqrt{d})$ and $\mathbb{Q}(\zeta_n)$.
- Frobenius for quadratic extensions: $\text{Frob}_p = (d/p)$; QR as the symmetry Frob_p in $\mathbb{Q}(\sqrt{q}) \leftrightarrow \text{Frob}_q$ in $\mathbb{Q}(\sqrt{p})$.

- Kronecker–Weber for quadratic extensions (proof via cyclotomic fields).
- The Legendre symbol re-read as “evaluation at the point q ”.

Main results proved. QR as Frobenius reciprocity; Kronecker–Weber (quadratic case).

Role in the course. *Depends on S5–S7. Feeds S9, S20, S46, Act II.* *Arc:* introduces the Frobenius element, central to Acts III & VI and the Langlands outlook.

Unifying thread. *Legendre = Frobenius = evaluation at a point, now group-valued.*

References. Ireland–Rosen ch.13; Neukirch ch.I §9.

3.5 Session 9 — Cyclotomic Extensions and Frobenius Elements

Content to cover.

- Cyclotomic polynomials; $\mathbb{Q}(\zeta_n)$; $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n)^\times$ (proof).
- Concrete decomposition of primes: split / inert / ramified according to $p \bmod n$; Dedekind’s criterion examples.
- Concrete inertia & ramification in cyclotomic fields.
- Signposting: valuation-theoretic e, f in S20; decomposition/inertia groups in S46.

Main results proved. Structure of cyclotomic Galois groups; concrete prime decomposition.

Role in the course. *Depends on S8. Feeds S16, S47–S49. Arc:* first infinite family where Frobenius is explicitly computable.

Unifying thread. *In $\mathbb{Q}(\zeta_n)$, the point p is completely described by $p \bmod n$.*

References. Lang, *ANT* ch.I; Ireland–Rosen ch.13.

3.6 Session 10 — Summary: From Reciprocity to Galois Theory

Content to cover.

- Recap of Act I; the hierarchy of reciprocity laws; the question of non-abelian generalizations.
- Preview of Act II as the language for those generalizations.

Role in the course. *Depends on S5–S9. Feeds Act II. Arc:* consolidation; sets the “generalize QR” agenda.

References. Review of Act I sources.

4 Act II — Galois Theory and Solvability

4.1 Session 11 — Solvable Groups, S_n , and A_n

Content to cover.

- S_n : cycle decomposition, sign homomorphism; $A_n = \ker(\text{sign})$.
- Simplicity of A_n for $n \geq 5$ (statement & proof idea via 3-cycles).
- Solvable groups via composition series with abelian quotients; commutator subgroup; S_3, S_4 solvable, S_5 not.

Main results proved. Simplicity of A_n ($n \geq 5$); solvability of S_3, S_4 ; non-solvability of S_5 .

Role in the course. *Depends on P1. Feeds S14, S15. Arc:* the engine turning symmetry into a solvability obstruction.

Unifying thread. *Composition factors are the elementary symmetries a point-system can have.*

References. Artin ch.2 §7–8; Rotman ch.4.

4.2 Session 12 — Galois Extensions, Finite Fields, and Linear Independence

Content to cover.

- Separable polynomials & extensions; normal extensions; splitting fields; characterizations of Galois extensions (proofs).
- Finite fields $\mathbb{F}_{p^n}$: existence, uniqueness, Galois group cyclic generated by Frobenius (proof).
- Dedekind's lemma: linear independence of characters (proof); trace & norm preview; used in S13, S14, S20, S45.

Main results proved. Galois characterizations; structure of finite fields; Dedekind's lemma.

Role in the course. *Depends on* S3, S11. *Feeds* S13, S14, S20, S45. *Arc:* finite-field Frobenius is the prototype of Frob_p (S46).

Unifying thread. *Frobenius is the canonical point-moving symmetry of finite fields.*

References. Lang ch.VI; Artin, *Galois Theory*; Conrad, *FGT*.

4.3 Session 13 — The Fundamental Theorem of Galois Theory

Content to cover.

- FTGT: inclusion-reversing bijection subgroups $\leftrightarrow$ intermediate fields (complete proof using S12).
- Examples $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$, $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$; normal subgroups $\leftrightarrow$ Galois subextensions.
- Primitive element theorem (statement/proof idea); Galois group of finite fields.

Main results proved. FTGT; normal $\leftrightarrow$ Galois.

Role in the course. *Depends on* S12. *Feeds* S14–S16, S35, S56. *Arc:* the dictionary later categorified (covers $\leftrightarrow \pi_1$ -sets) and logicalized (subtopos $\leftrightarrow$ theory quotient).

Unifying thread. *Intermediate fields are subspaces of points of the Galois correspondence.*

References. Artin ch.4; Lang ch.VI.

4.4 Session 14 — Solvable Groups and Solvability by Radicals

Content to cover.

- Radical extensions; Kummer theory (statement); roots of unity.
- Theorem: f solvable by radicals $\iff \text{Gal}(f)$ solvable (proof using S11–S13).
- Worked examples of solvable and non-solvable polynomials.

Main results proved. Solvability-by-radicals characterization.

Role in the course. *Depends on* S11–S13. *Feeds* S15. *Arc:* the classical payoff of the Galois dictionary.

References. Artin ch.5; Lang ch.VI §8.

4.5 Session 15 — The Insolvability of the Quintic

Content to cover.

- The general polynomial of degree n ; its Galois group S_n .
- Abel–Ruffini: degree ≥ 5 not solvable by radicals (proof via S11).
- An explicit unsolvable quintic (example/mention).

Main results proved. Abel–Ruffini.

Role in the course. *Depends on* S14. *Feeds* S16. *Arc:* closes the classical problem; motivates studying Galois groups as objects.

References. Artin ch.5; Stewart ch.3–4.

4.6 Session 16 — The Absolute Galois Group

Content to cover.

- $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$; Krull/profinite topology as an inverse limit (S4).
- The inverse Galois problem; why $G_{\mathbb{Q}}$ is mysterious; local decomposition preview (S46).
- Abelianization $G_{\mathbb{Q}}^{\text{ab}}$; Kronecker–Weber (statement); preview of CFT (Act VI).
- $G_{\mathbb{Q}}$ as “fundamental group of $\text{Spec } \mathbb{Q}$ ” (made precise S35).

Main results proved. Profinite structure of $G_{\mathbb{Q}}$; Kronecker–Weber stated & checked on examples.

Role in the course. *Depends on* S4, S13. *Feeds* Acts III, VI, VII. *Arc:* the master symmetry space of all algebraic points over $\mathbb{Q}$.

Unifying thread. $G_{\mathbb{Q}}$ is the holonomy of the covering $\text{Spec } \bar{\mathbb{Q}} \rightarrow \text{Spec } \mathbb{Q}$.

References. Neukirch ch.I; Szamuely ch.1.

5 Act III — Valuations and p -adic Points

5.1 Session 17 — Absolute Values, Valuations, and Ostrowski’s Theorem

Content to cover.

- Absolute values; equivalence; archimedean vs non-archimedean; additive valuations $v : K^* \rightarrow \Gamma$.
- Valuation ring $\mathcal{O}_v$, maximal ideal $\mathfrak{m}_v$, residue field k_v , value group.
- Ostrowski’s theorem for $\mathbb{Q}$ (proof sketch): places of $\mathbb{Q}$ are $\{\infty, 2, 3, 5, \dots\}$.
- Remark: Ostrowski for $k(t)$ (function-field places) previewing S21.

Main results proved. Ostrowski (sketch); basic properties of valuation rings.

Role in the course. *Depends on* P2. *Feeds* S18–S22, S33, S46. *Arc:* the census of points of $\mathbb{Q}$.

Unifying thread. *A place is a point; Ostrowski lists every point of $\mathbb{Q}$.*

References. Gouvêa ch.1–2; Neukirch ch.II §1.

5.2 Session 18 — Construction of $\mathbb{Q}_p$ and $\mathbb{Z}_p$

Content to cover.

- Completion of $\mathbb{Q}$ w.r.t. $|\cdot|_p$ via the P2 machine; $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n$ (S4); equivalence of the two constructions.
- Series representation $\sum a_i p^i$; valuation & topology on $\mathbb{Q}_p$; $\mathbb{Z}_p$ a local ring with residue $\mathbb{F}_p$.
- Unit structure $\mathbb{Q}_p^* \cong p^{\mathbb{Z}} \times \mathbb{Z}_p^*$; concrete computations ($-1, 1/2, \sqrt{7}$ in $\mathbb{Z}_3$).
- $\mathbb{Q}_p$ not algebraically closed; $\bar{\mathbb{Q}}_p$ and $\mathbb{C}_p$ (mention).

Main results proved. Equivalence of completion & inverse-limit constructions; unit structure.

Role in the course. *Depends on* P2, S4, S17. *Feeds* S19, S22, S34, S50, S28/S30. *Arc:* $\mathbb{Q}$ completed at the point p .

Unifying thread. *Completion at a point yields a new, purely local number system.*

References. Gouvêa ch.3–4; Serre, *Course in Arithmetic* ch.II.

5.3 Session 19 — Hensel’s Lemma and p -adic Topology

Content to cover.

- Hensel’s lemma (proof via Newton iteration/contraction); applications to roots & factorization ($\sqrt{2}$, $\sqrt{-1}$ in $\mathbb{Q}_p$?).
- The ultrametric inequality; all triangles isosceles; open balls clopen.
- Compactness of $\mathbb{Z}_p$ (proof); local compactness of $\mathbb{Q}_p$.
- Hensel as “smooth points lift”: preview of étale (S30).

Main results proved. Hensel’s lemma; compactness of $\mathbb{Z}_p$.

Role in the course. *Depends on* P2, S18. *Feeds* S22, S30, S34. *Arc:* the local inverse-function theorem of arithmetic.

Unifying thread. *Hensel lifts residue-field points to genuine local points.*

References. Gouvêa ch.4; Neukirch ch.II §4.

5.4 Session 20 — Extension of Valuations: Ramification and Residue Degree

Content to cover.

- Extending v from K to L ; ramification index e ; residue degree f (the formal definitions promised in S9).
- Fundamental inequality $\sum e_i f_i \leq [L : K]$ (proof via Dedekind’s lemma, S12); equality in the Galois case.
- Unramified / split / inert; examples in quadratic & cyclotomic fields; relation to S9.
- Preview of decomposition & inertia groups (S46).

Main results proved. Fundamental inequality; equality for Galois extensions.

Role in the course. *Depends on* S12, S17. *Feeds* S24, S46, S50. *Arc:* makes S9’s concrete decomposition rigorous.

Unifying thread. *Ramification measures how several points merge into one.*

References. Neukirch ch.I §8–9; Lang, *ANT* ch.I.

5.5 Session 21 — Function Fields and the Geometric Analogy

Content to cover.

- $k(t)$ and its places = points of $\mathbb{P}^1$; the t -adic valuation; completion $k((t))$.
- The analogy table: $\mathbb{Q}_p \leftrightarrow k((t))$, $\mathbb{Z}_p \leftrightarrow k[[t]]$, prime $p \leftrightarrow$ point $t = 0$.
- Divisors & degree preview; Riemann–Roch preview (S32, S38).
- Remark: the number-field / function-field dictionary as a working philosophy.

Main results proved. Place–point correspondence for $k(t)$; completion to $k((t))$.

Role in the course. *Depends on* S17. *Feeds* S32, S36, S38, S60. *Arc:* makes explicit the identification “prime $\leftrightarrow$ point”.

Unifying thread. $\text{Spec } \mathbb{Z}$ and a curve are the same kind of object.

References. Gouvêa ch.2; Rosen ch.1.

5.6 Session 22 — Adeles, Ideles, and Adelic Compactness

Content to cover.

- Restricted product: definition, topology, local compactness (using P2); the adèle ring $\mathbb{A}_{\mathbb{Q}}$ & idele group $\mathbb{A}_{\mathbb{Q}}^*$.
- $\mathbb{Q}$ embedded diagonally & discrete; Product Formula $\prod_v |x|_v = 1$ (proof).
- $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is compact (proof sketch via a fundamental domain); consequences previewed (S24).
- Adelic viewpoint on ideals & class groups (preview S24); automorphic preview (S40, S61).

Main results proved. Product Formula; adelic compactness (sketch).

Role in the course. *Depends on* P2, S18. *Feeds* S24, S41, S49, S61. *Arc:* adeles = all points simultaneously; compactness = the finiteness engine.

Unifying thread. *The adèle ring is tuples of local points glued by the product formula.*

References. Neukirch ch.VII §1; Cassels–Fröhlich ch.VII; Weil.

6 Act IV — Algebraic Geometry

6.1 Session 23 — Commutative Algebra I

Content to cover.

- Noetherian rings & equivalent conditions; Hilbert Basis Theorem (proof).
- Integral elements; integral closure; integrally closed domains; examples.
- Krull dimension; dimension of local rings; examples $\mathcal{O}_K$, coordinate rings, $\mathbb{Z}$.
- Local rings & completion (preview of the S18/S30 interplay).

Main results proved. Hilbert Basis; integral-closure properties; dimension computations.

Role in the course. *Depends on* S1–S2. *Feeds* S24, S25, S28, S30. *Arc:* finiteness & integrality make arithmetic points behave geometrically.

Unifying thread. *Dimension counts independent directions in which points can move.*

References. Atiyah–Macdonald ch.5,7; Eisenbud ch.4,7.

6.2 Session 24 — Dedekind Domains and Ideal Class Groups

Content to cover.

- Dedekind domains & equivalent characterizations; unique factorization of ideals (proof).
- Fractional ideals & their group; the class group $\text{Cl}(\mathcal{O}_K)$; example $\text{Cl}(\mathbb{Z}[\sqrt{-5}]) \cong \mathbb{Z}/2$.
- Finiteness of the class group via adelic compactness (S22) (proof); Minkowski alternative (mention).
- Relation to ideal factorization (S20); bridge to S46–S47; recalled again at S46.

Main results proved. Unique factorization of ideals; finiteness of the class group (adelic route).

Role in the course. *Depends on* S20, S22, S23. *Feeds* S46, S47–S49. *Arc:* the arithmetic of number rings placed inside the geometry act.

Unifying thread. *The class group measures how far the points of $\mathrm{Spec} \mathcal{O}_K$ are from being principal.*

References. Neukirch ch.I §3; Lang, *ANT* ch.I; Cassels–Fröhlich.

6.3 Session 25 — Affine Varieties and Hilbert’s Nullstellensatz

Content to cover.

- Algebraic sets in k^n ; the Zariski topology on k^n ; coordinate rings.
- Hilbert’s Nullstellensatz (proof via Rabinowitch / Noether normalization); algebraic-set $\leftrightarrow$ radical-ideal correspondence.
- Morphisms of affine varieties; maximal ideals as closed points (over $\bar{k}$).

Main results proved. Nullstellensatz; the classical correspondence.

Role in the course. *Depends on* S23. *Feeds* S28, S31. *Arc:* the classical bridge where “point = maximal ideal” first becomes a theorem.

Unifying thread. *Over an algebraically closed field, geometry is recovered from maximal ideals.*

References. Vakil ch.3; Hartshorne ch.I.

6.4 Session 26 — Presheaves and Sheaves

Content to cover.

- Presheaf as contravariant functor $\mathrm{Open}(X)^{\mathrm{op}} \rightarrow \mathcal{C}$; restriction maps; sections; $\mathrm{PSh}(X)$.
- Examples & non-examples: continuous; holomorphic (defined inline; full theory S37–S38); constant presheaf not a sheaf; bounded functions fail gluing.
- Sheaf axioms (uniqueness + gluing); equalizer-diagram formulation; sheaf morphisms; kernel & cokernel.
- $\mathrm{Epi} \neq$ surjective on sections: the $\exp : \mathcal{O} \rightarrow \mathcal{O}^*$ example on $\mathbb{C}$.
- Stalks as directed colimits (S4); germs of sections.

Main results proved. Equalizer formulation equivalent to gluing axioms; kernel is a sheaf; the $\exp$ counterexample.

Role in the course. *Depends on* S4, P2. *Feeds* S27, S28, S53, S54. *Arc:* a sheaf assigns data to every open and reconciles local into global.

Unifying thread. *The stalk at x is the value of the sheaf at the point x .*

References. Vakil ch.2; Hartshorne ch.II §1; Tennison; Mac Lane–Moerdijk.

6.5 Session 27 — Sheafification, Categories of Sheaves, and Étale Spaces

Content to cover.

- Sheafification & universal property; plus-construction $\mathcal{F} \rightarrow \mathcal{F}^+ \rightarrow \mathcal{F}^{++}$ (the idempotent-monad remark of S4 made concrete).
- Adjunction $a \dashv i$; $\mathrm{Sh}(X)$ as reflective subcategory / localization of $\mathrm{PSh}(X)$; limits & colimits in $\mathrm{Sh}(X)$.
- Direct image f_* & inverse image f^{-1} ; adjunction $f^{-1} \dashv f_*$.

- Étale-space equivalence: $\mathrm{Sh}(X) \simeq$ spaces over X via local homeomorphisms; stalk = fiber $\pi^{-1}(x)$; skyscraper & constant sheaves via étale spaces.
- Origin of the word “étale” and its reuse in S30 / S35 / S52.

Main results proved. Sheafification universal property; $f^{-1} \dashv f_*$; the étale-space equivalence (construction).

Role in the course. *Depends on* S26. *Feeds* S28, S53, S35/S52. *Arc:* a sheaf is a space of germs; its points are its stalks.

Unifying thread. *Sheafification turns inconsistent local data into a genuine space of points.*

References. Mac Lane–Moerdijk ch.II,VI; Tennison.

6.6 Session 28 — Schemes: $\mathrm{Spec}(R)$, the Structure Sheaf, Closed & Generic Points

Content to cover.

- Locally ringed spaces; affine schemes; structure sheaf $\mathcal{O}_{\mathrm{Spec} R}$ (glueing proof); stalks $R_{\mathfrak{p}}$; residue fields $\kappa(\mathfrak{p})$.
- Closed points = maximal ideals; generic points = minimal primes; specialization; examples $\mathrm{Spec} \mathbb{Z}$, $\mathbb{A}^1$, a curve.
- Morphisms of schemes; first examples; the non-Hausdorff nature revisited.

Main results proved. Structure presheaf is a sheaf; stalk & residue computations.

Role in the course. *Depends on* S1, S26–S27. *Feeds* S29–S35, S46, S54. *Arc:* the stage on which points are primes with local rings of germs.

Unifying thread. *A scheme is a space whose points carry their own local algebra.*

References. Vakil ch.4; Hartshorne ch.II §2; Görtz–Wedhorn ch.3.

6.7 Session 29 — The Functor of Points and Yoneda’s Perspective

Content to cover.

- Yoneda lemma (complete proof); representable functors; the Yoneda embedding.
- Functor of points $h_X(T) = \mathrm{Hom}(T, X)$; T -valued points; examples $(X(T))$ as solution sets of equations; $X \cong \mathrm{Hom}(-, X)$.
- A scheme is determined by its functor of points; natural transformations = morphisms.
- Non-representable functors $\rightarrow$ moduli problems (preview S41).

Main results proved. Yoneda lemma.

Role in the course. *Depends on* S4, S28. *Feeds* S41, S54, S56. *Arc:* the conceptual heart: point = morphism into the object.

Unifying thread. *To know a space is to know its points over every possible base.*

References. Vakil ch.7,9; Eisenbud–Harris ch.VI; Demazure–Gabriel ch.1.

6.8 Session 30 — Commutative Algebra II: Morphisms, Fiber Products, Étale Maps

Content to cover.

- Morphisms of schemes; fiber products via tensor products (S2); base change & fibers.

- Properties: finite type, separated; flat (serious treatment now, previewed S2), unramified, étale, smooth.
- Jacobian criterion (proof idea); examples & non-examples.
- Étale = “algebraic local homeomorphism”: Hensel-lifting characterization (S19).

Main results proved. Existence of fiber products of affine schemes; Jacobian criterion.

Role in the course. *Depends on* S2, S28. *Feeds* S33, S35, S39, S52. *Arc:* maps that preserve local point-structure.

Unifying thread. *Étale maps are the maps under which points look locally identical.*

References. Görtz–Wedhorn ch.6,11; Vakil ch.7,13; Stacks Project.

6.9 Session 31 — Projective Varieties and Points at Infinity

Content to cover.

- Graded rings & homogeneous ideals; $\text{Proj}(S)$; $\mathbb{P}^n$; affine charts.
- Points at infinity & why they are necessary; projective = proper (preview S33).
- Bézout’s theorem (statement); examples of intersection at infinity.

Main results proved. Covering by affine charts; basic properties of Proj .

Role in the course. *Depends on* S28. *Feeds* S32, S39, S41. *Arc:* completeness: adding the points at infinity makes every count exact.

Unifying thread. *Projective space is affine space with its missing points restored.*

References. Vakil ch.6; Hartshorne ch.I §2.

6.10 Session 32 — Divisors, the Product Formula, and Riemann–Roch (Algebraic)

Content to cover.

- Divisors $\sum n_P[P]$; principal divisors $(f) = \sum v_P(f)[P]$; $\deg(f) = 0$ on a projective curve (proof).
- Arithmetic analogue: the product formula (S22); function-field analogue (S21).
- Canonical divisor & Kähler differentials; genus; Riemann–Roch $\ell(D) - \ell(K - D) = \deg D + 1 - g$ (algebraic proof via Riemann’s theorem).
- Examples $\mathbb{P}^1$, elliptic curves; Remark: the analytic viewpoint in S38.

Main results proved. $\deg(f) = 0$; Riemann–Roch for curves (algebraic).

Role in the course. *Depends on* S21, S31. *Feeds* S39, S38, S41. *Arc:* the exact accounting of zeros & poles across all points.

Unifying thread. *Riemann–Roch balances the vanishing of functions at every point of the curve.*

References. Hartshorne ch.IV; Fulton; Miranda (analytic companion).

6.11 Session 33 — The Valuative Criterion of Properness

Content to cover.

- Separated & proper morphisms (definitions using S30 fiber products).
- Valuative criterion of properness (statement; proof for curves); restatement via specialization generic $\rightarrow$ closed.

- Examples: $\mathbb{A}^1$ not proper, $\mathbb{P}^1$ proper; analogy with compactness (S22).

Main results proved. Valuative criterion (curves).

Role in the course. *Depends on* S20, S30, S31. *Feeds* S41, S36. *Arc:* makes “complete = no missing points” rigorous.

Unifying thread. *Properness is the refusal of points to run away to infinity.*

References. Vakil ch.7; Hartshorne ch.II §4.

6.12 Session 34 — Free Groups and Profinite Groups

Content to cover.

- Free groups F_n : universal property; examples; $F_1 \cong \mathbb{Z}$.
- Profinite groups as inverse limits of finite groups (S4); $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n$; profinite topology compact & totally disconnected (P2).
- Continuous actions of profinite groups; Galois groups as profinite (S16).

Main results proved. Universal property of free groups; compactness of profinite groups.

Role in the course. *Depends on* P2, S4, S16. *Feeds* S35, S59. *Arc:* the group objects in which $\pi_1^{\text{ét}}$ and $G_{\mathbb{Q}}$ live.

Unifying thread. *A profinite group is the Galois group of an infinite system of points.*

References. Ribes–Zalesskii ch.1; Szamuely ch.1; Serre, *Galois Cohomology* ch.I.

6.13 Session 35 — Étale Fundamental Group and Grothendieck’s Galois Theory

Content to cover.

- Étale covers (S30); the fiber functor; definition of $\pi_1^{\text{ét}}(X, \bar{x})$.
- Grothendieck’s correspondence: finite étale covers $\leftrightarrow$ finite continuous $\pi_1^{\text{ét}}$ -sets (statement & construction), generalizing the topological theory of P3.
- Case $X = \text{Spec}(K)$ recovers FTGT (S13); examples $\pi_1^{\text{ét}}(\mathbb{G}_{m,\mathbb{C}}) = \hat{\mathbb{Z}}$, $\pi_1^{\text{ét}}(\mathbb{P}^1 \setminus \{0, 1, \infty\}) = \hat{F}_2$.
- Using the topological intuition of P3 (coverings $\leftrightarrow \pi_1$); preview of Dessins (S60).

Theorem (Grothendieck). *Finite étale covers of a connected X with geometric point $\bar{x} \leftrightarrow$ finite sets with a continuous $\pi_1^{\text{ét}}(X, \bar{x})$ -action.*

Main results proved. Grothendieck’s correspondence (construction); the $\text{Spec } K$ case.

Role in the course. *Depends on* P3, S13, S30, S34. *Feeds* S52–S53, S59, S60. *Arc:* unifies topology, Galois theory, and étale geometry.

Unifying thread. $\pi_1^{\text{ét}}$ is the symmetry group of all finite point-systems over X .

References. Szamuely ch.4–5; SGA1 exp.V.

6.14 Session 36 — Summary: The Bridge Between Number Theory and Geometry

Content to cover.

- $\text{Spec } \mathbb{Z}$ as a “curve”; primes = points; ∞ = the point at infinity.
- The big number-field / function-field analogy table; CFT previewed as abelian Galois theory of $\text{Spec } \mathbb{Z}$.

Role in the course. *Depends on Acts III–IV. Feeds Acts V–VI. Arc:* consolidation; sets the CFT agenda.

References. Lorenzini; review of Acts III–IV.

7 Act V — Curves over $\mathbb{C}$ and Elliptic Curves

7.1 Session 37 — Complex Analysis I: Holomorphic Functions & Cauchy Theory

Content to cover.

- Holomorphic functions; Cauchy–Riemann equations; power-series expansion.
- Cauchy’s theorem & integral formula; identity theorem; Liouville; maximum principle.
- Meromorphic functions; residues & the argument principle (mention).

Main results proved. Cauchy integral formula; identity theorem; Liouville.

Role in the course. *Depends on P2. Feeds S38, S40, and the S26 holomorphic example. Arc:* analytic continuation = global from local.

Unifying thread. *A holomorphic function is rigid local data: its germ at one point knows everything.*

References. Ahlfors ch.1–4; Conway.

7.2 Session 38 — Complex Analysis II: Riemann Surfaces

Content to cover.

- Riemann surfaces: definition & examples; meromorphic functions & their divisors; $\deg$ of a principal divisor = 0 (proof) — the analytic product formula.
- Compact Riemann surfaces = smooth projective curves over $\mathbb{C}$ (GAGA flavour); genus.
- Analytic Riemann–Roch (statement), completing S32’s remark; uniformization (mention).
- Complex tori $\mathbb{C}/\Lambda$; Weierstrass $\wp$; tori $\leftrightarrow$ elliptic curves; upper half-plane $\mathbb{H}$ & $\mathrm{SL}_2(\mathbb{Z})$ action (preview S40–S41).

Main results proved. Analytic product formula; torus = elliptic curve (construction).

Role in the course. *Depends on S37, S32. Feeds S39, S40, S41. Arc:* the analytic avatar of algebraic curves; topological genus = arithmetic genus.

Unifying thread. *A compact Riemann surface is a curve; its points are the places of its function field.*

References. Forster; Miranda; Serre, *GAGA* (mention).

7.3 Session 39 — Elliptic Curves: Geometry and Group Law

Content to cover.

- Weierstrass equations; smoothness via the Jacobian criterion (S30); discriminant.
- Projective closure; the point at infinity $\mathcal{O}$ as the identity element.
- Chord-and-tangent law; group axioms via Riemann–Roch (S32) (proof); associativity; torsion points.
- Over $\mathbb{C}$: $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$ (S38); torus addition = chord-and-tangent.

Main results proved. Group law well-defined & associative (via RR).

Role in the course. *Depends on* S30, S32, S38. *Feeds* S40, S41. *Arc:* the central example; its identity is literally a point at infinity.

Unifying thread. *The group structure lives on the points; the identity is the point at infinity.*

References. Silverman ch.I–III; Washington.

7.4 Session 40 — Elliptic Curves: Arithmetic and Modular Forms

Content to cover.

- Reduction mod p : good / multiplicative / additive; the L -function of an elliptic curve.
- Mordell–Weil theorem (statement).
- Modular forms on $\mathbb{H}$ (holomorphy from S37–S38): weight- k , growth at cusps; Hecke operators (mention).
- Modularity Theorem (statement); the link to Fermat’s Last Theorem.

Main results proved. Basic reduction properties; modularity stated & verified on examples.

Role in the course. *Depends on* S37–S39. *Feeds* S41, S61. *Arc:* the first non-abelian-flavoured reciprocity; L -functions package point-counts.

Unifying thread. *The L -function packages the point-counts of $E \bmod p$ into one analytic object.*

References. Silverman ch.VII–VIII; Diamond–Shurman ch.1–3.

7.5 Session 41 — Points as Objects: Moduli Spaces and Rational Points

Content to cover.

- The j -invariant; the moduli problem $\mathcal{M}_{1,1}$: points = isomorphism classes of elliptic curves.
- (Non-)representability & the need for stacks (survey); compactified moduli; ties to S29.
- Rational points & Diophantine geometry: Mordell–Weil, Faltings (statements); BSD (mention).

Main results proved. j -invariant classifies elliptic curves over algebraically closed fields.

Role in the course. *Depends on* S29, S39–S40. *Feeds* S56, S60–S61. *Arc:* objects become points; arithmetic asks which points are rational.

Unifying thread. *Moduli spaces turn objects into points; arithmetic asks which points are rational.*

References. Silverman ch.III; Hartshorne ch.IV; Mazur; Katz–Mazur (selected); Hindry–Silverman (survey).

8 Act VI — Class Field Theory

8.1 Session 42 — Homological Algebra I: Complexes and the Snake Lemma

Content to cover.

- Chain complexes; chain maps; homology; exact sequences; abelian categories (mention).
- Snake lemma (proof / proof idea); five lemma; the long exact sequence of homology.

Main results proved. Snake lemma; five lemma; long exact sequence.

Role in the course. *Depends on* S2. *Feeds* S43–S45. *Arc:* the homological engine that makes reciprocity provable.

References. Weibel ch.1–2; Rotman, *Homological Algebra* ch.1–2.

8.2 Session 43 — Homological Algebra II: Resolutions and Derived Functors

Content to cover.

- Projective & injective modules; resolutions; derived functors.
- Tor & Ext; Ext^1 classifies extensions (proof idea); long exact sequences of derived functors.
- Group cohomology as Ext over $\mathbb{Z}G$ (preview S44); uses S2.

Main results proved. Existence of resolutions; Ext^1 classification; derived long exact sequences.

Role in the course. *Depends on* S42. *Feeds* S44, S45, S59. *Arc:* the derived language in which reciprocity is an H^2 statement.

References. Weibel ch.2–3; Mac Lane, *Homology*.

8.3 Session 44 — Group Cohomology and Tate Cohomology

Content to cover.

- G -modules; invariants; the cochain complex $C^n(G, M)$; H^0, H^1, H^2 explicitly.
- H^1 = crossed homomorphisms mod principal; H^2 = group extensions.
- Derived-functor viewpoint $H^n(G, M) = \text{Ext}_{\mathbb{Z}G}^n(\mathbb{Z}, M)$ (S43).
- Tate cohomology $\hat{H}^n$ for finite G ; periodicity for cyclic groups (proof idea).

Main results proved. Equivalence of cochain & derived definitions in low degrees; Tate periodicity for cyclic groups.

Role in the course. *Depends on* S43. *Feeds* S45, S47. *Arc:* the cohomology of symmetry groups, ready for Galois groups.

References. Serre, *Galois Cohomology* ch.I; Cassels–Fröhlich ch.VIII; Weibel ch.6.

8.4 Session 45 — Galois Cohomology and the Fundamental Exact Sequence

Content to cover.

- Galois cohomology of L^* ; Hilbert’s Theorem 90 (proof via Dedekind’s lemma, S12).
- The Brauer group (mention); local invariant $H^2(\mathbb{Q}_p) \cong \mathbb{Q}/\mathbb{Z}$ (statement).
- The fundamental exact sequence $0 \rightarrow H^2 \rightarrow \bigoplus_v \mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ (proof using S42 snake lemma & S22 product formula).
- The cohomological statement of class field theory (preview S47).

Main results proved. Hilbert 90; local invariant isomorphism; the fundamental exact sequence.

Role in the course. *Depends on* S22, S44. *Feeds* S47, S49. *Arc:* the product formula rewritten in cohomology.

Unifying thread. *The exact sequence is the cohomological shadow of “the product of all local evaluations is 1”.*

References. Serre ch.II; Neukirch ch.IV.

8.5 Session 46 — The Artin Map

Content to cover.

- Recap from S24: fractional ideals & the class group, freshened after the gap.
- Decomposition & inertia groups (formal definitions, as promised in S9); residue field extensions; Frobenius conjugacy class.

- Unramified primes give a well-defined Frobenius element; the Artin symbol $(L/K/\mathfrak{p})$; extension to fractional ideals.
- Examples: cyclotomic & quadratic extensions.

Main results proved. Well-definedness of Frobenius; the Artin map is a homomorphism on ideals.

Role in the course. *Depends on* S9, S20, S24. *Feeds* S47–S49. *Arc:* evaluation at a point, now group-valued & multiplicative.

Unifying thread. *The Artin map evaluates the prime $\mathfrak{p}$ into the symmetry group of the extension.*

References. Neukirch ch.VII §3; Lang, *ANT* ch.XI.

8.6 Session 47 — Artin Reciprocity

Content to cover.

- Artin Reciprocity Law: statement; kernel = norm group; $\text{Gal}(L/K) \cong I_K/N_{L/K}(I_L) \cdot P_K$.
- Complete proofs for quadratic & cyclotomic extensions; general outline via S45.
- Quadratic reciprocity recovered as a corollary (S6).

Main results proved. Artin reciprocity for quadratic & cyclotomic; general outline.

Role in the course. *Depends on* S45, S46. *Feeds* S48–S49, S61. *Arc:* the First Isomorphism Theorem (P1) realized at global scale.

Unifying thread. *Global reciprocity: the product of all local evaluations is the identity.*

References. Neukirch ch.VII §6; Cassels–Fröhlich ch.X.

8.7 Session 48 — The Existence Theorem

Content to cover.

- Existence Theorem (Takagi–Artin): statement; open subgroups of finite index $\leftrightarrow$ abelian extensions.
- Proof ideas (cohomological / adelic); the Hilbert class field; examples.

Main results proved. Existence theorem (ideas & structure).

Role in the course. *Depends on* S47. *Feeds* S49. *Arc:* every admissible subgroup of “points” comes from a field.

References. Neukirch ch.VII §8; Lang ch.XIII.

8.8 Session 49 — The Idele Formulation and Global CFT

Content to cover.

- Ideles J_K & idele class group C_K (S22); the Artin map in idele language.
- Reciprocity: $C_K/N(C_L) \cong \text{Gal}(L/K)$; $G_K^{\text{ab}} \cong C_K$ (statement).
- Comparison with the ideal version; the role of adelic compactness (S22).

Main results proved. Idele-theoretic reciprocity; the abelianization isomorphism.

Role in the course. *Depends on* S22, S47–S48. *Feeds* S61. *Arc:* the modern formulation; C_K = product of all local points modulo globals.

Unifying thread. *Global abelian symmetry is exactly the idele class group — all points at once.*

References. Neukirch ch.VII; Cassels–Fröhlich ch.XII.

8.9 Session 50 — Local Class Field Theory: Lubin–Tate

Content to cover.

- Formal group laws (self-contained opening): 1-dim commutative FGLs over a ring; additive & multiplicative examples; formal logarithm; the notion of height.
- Lubin–Tate series & formal modules; Theorem: all abelian extensions of $\mathbb{Q}_p$ arise from Lubin–Tate theory (key ideas).
- The local reciprocity map $\mathbb{Q}_p^* \rightarrow \text{Gal}(L/\mathbb{Q}_p)^{\text{ab}}$; the local analogue of Kronecker–Weber.

Main results proved. Lubin–Tate theorem (key ideas); local reciprocity.

Role in the course. *Depends on* S18–S19, S44. *Feeds* S61. *Arc:* local CFT made explicit; formal groups = points near the identity.

Unifying thread. *Local reciprocity: the prime’s own formal group generates all its abelian symmetry.*

References. Serre, *Local Fields* ch.XIV–XV; Neukirch ch.V §6; Iwasawa.

9 Interlude — Spaces without Points

9.1 Session 51 — Gelfand–Naimark and the Non-commutative Glimpse

Content to cover.

- C^* -algebras: minimal definitions; the Gelfand transform.
- Theorem (Gelfand–Naimark, categorical statement): commutative unital C^* -algebras $\simeq$ compact Hausdorff spaces, contravariantly (via maximal ideals = “points”); no functional analysis assumed.
- Points of X = characters of $C(X)$ = maximal ideals (one more appearance).
- Non-commutative C^* -algebra = “space without classical points” (Connes perspective, survey).
- Slogan: dropping commutativity drops points; but the algebra still remembers geometry.
- Scope note: no spectral theory, no dagger categories — pointers as further reading only; echoes S1/S28 and previews Act VII.

Main results proved. Gelfand–Naimark duality (categorical statement & proof outline).

Role in the course. *Depends on* S1, P2. *Feeds* S55, S58, S61. *Arc:* shows “space without points” is meaningful, motivating the logical treatment of points.

Unifying thread. *Dropping commutativity drops classical points — but the algebra still remembers the geometry.*

References. Gracia-Bondía–Várilly–Figueroa ch.3 (survey); Kadison–Ringrose ch.4; Connes (further reading).

10 Act VII — Topos Theory and Logic

10.1 Session 52 — Sites and Grothendieck Topologies

Content to cover.

- Grothendieck topologies: coverings & sieves; sites $(\mathcal{C}, J)$.
- Examples: Zariski, étale, fppf; contrast with point-set topology (P2, S28).
- Subcanonical sites; the Yoneda embedding of a site (S29).

Main results proved. Equivalence of sieve & covering-family formulations.

Role in the course. *Depends on S4, S28, S30. Feeds S53, S59. Arc:* generalizes “open cover” so point-less spaces still have sheaf theory.

Unifying thread. *A covering says how points cover — even when the points are not there.*

References. Mac Lane–Moerdijk ch.III; Johnstone ch.C.

10.2 Session 53 — Sheaves on Sites and Topoi

Content to cover.

- Sheaves on a site (generalizing S26–S27); the sheaf condition for arbitrary covers.
- A topos = $\text{Sh}(\mathcal{C}, J)$; examples $\text{Sh}(X)$, X_{Zar} , $X_{\text{ét}}$, G -sets.
- The generalized plus-construction (S27); limits/colimits; subobject classifier (mention).

Main results proved. Sheaf-condition equivalences; topos completeness/cocompleteness.

Role in the course. *Depends on S27, S52. Feeds S54–S59. Arc:* the world in which S54’s points live.

Unifying thread. *A topos is “the world of sheaves” — a space whose points may be invisible.*

References. Mac Lane–Moerdijk ch.III; Johnstone ch.C.

10.3 Session 54 — Geometric Morphisms and Points of a Topos

Content to cover.

- Geometric morphisms (f^*, f_*) with $f^* \dashv f_*$ and f^* left exact.
- Definition of a point: $p : \mathbf{Set} \rightarrow \mathcal{E}$; the stalk functor p^* .
- Examples: point of a space $\rightarrow$ stalk; point of $\text{Spec } R \rightarrow$ prime $\rightarrow$ localization; étale point $\rightarrow$ geometric point $\text{Spec}(\Omega) \rightarrow X$; points of G -sets.
- Comparison with S29: point = morphism into the object.

Main results proved. Equivalence of point definitions (stalk-functor vs geometric morphism).

Role in the course. *Depends on S4, S29, S53. Feeds S55, S58. Arc:* the final, unifying definition of point.

Unifying thread. *Point = “morphism from $\mathbf{Set}$ into the topos” — evaluation from the outside.*

References. Mac Lane–Moerdijk ch.IV; Johnstone ch.C.2.

10.4 Session 55 — Enough Points

Content to cover.

- Conservative families of points: definition; examples & counterexamples.
- A topos with no points (explicit example); comparison with S51 (pointless spaces).
- Deligne’s theorem previewed (S58).

Main results proved. Basic properties of conservative families; the point-free example.

Role in the course. *Depends on S54. Feeds S58. Arc:* states the crisis — points can fail to exist — that Deligne resolves.

Unifying thread. *Enough points = “evaluations determine the structure”.*

References. Mac Lane–Moerdijk ch.IV §6; Johnstone ch.C.2.

10.5 Session 56 — Classifying Topoi and Geometric Theories

Content to cover.

- Geometric logic: signatures, geometric formulas & axioms, models (gentle introduction).
- The classifying topos $\mathbf{Set}[T]$ & its universal property; points of $\mathbf{Set}[T] =$ models of T in $\mathbf{Set}$.
- Examples: theory of fields, of groups, of graphs; comparison with S41 (moduli as classifying objects).

Main results proved. Points-of-classifying-topos = models (construction).

Role in the course. *Depends on S53. Feeds S57, S58. Arc:* a theory has a space whose points are its models.

Unifying thread. *The classifying topos is the space of models = the space of points of a theory.*

References. Mac Lane–Moerdijk ch.V; Johnstone ch.D.

10.6 Session 57 — Geometric Model Theory

Content to cover.

- Types & Stone spaces; saturated models (gentle introduction).
- Theorem: the type space of $T \cong$ the space of points of $\mathbf{Set}[T]$; saturated models = points realizing all types.
- Logical completeness $\leftrightarrow$ enough points.
- Pearl (optional, time-permitting): Lawvere’s fixed-point theorem — the categorical diagonal behind Cantor, Russell, Gödel, Tarski.

Main results proved. Type-space / point-space correspondence (statement & construction).

Role in the course. *Depends on S56. Feeds S58. Arc:* the logical reading of points; a type is a “generic point”.

Unifying thread. *Model theory is the algebraic geometry of theories; types are their points.*

References. Makkai–Reyes; Johnstone ch.D; Hodges; Lawvere (1969).

10.7 Session 58 — Deligne’s Theorem: Statement and Proof

Content to cover.

- Coherent topoi: definition & examples.
- Deligne’s Theorem: every coherent topos has enough points; full proof via Zorn (S1) + stepwise model construction.
- Equivalence with the completeness theorem for coherent geometric logic.

Theorem (Deligne, SGA4). *Every coherent topos has enough points; equivalently every consistent coherent geometric theory has a $\mathbf{Set}$ -model.*

Main results proved. Deligne’s theorem (full proof).

Role in the course. *Depends on S1, S55–S57. Feeds S59, S60. Arc:* the crown of the course: models are enough = points are enough.

Unifying thread. *Deligne: coherence forces a space of logic to have points.*

References. Mac Lane–Moerdijk ch.VII; SGA4 exp.VI; Johnstone ch.D.3.

10.8 Session 59 — Deligne’s Theorem in Algebraic Geometry

Content to cover.

- The étale topos of a qcqs scheme is coherent (S28, S52).
- Étale points = geometric points; étale sheaves determined by stalks at geometric points.
- Application to étale cohomology (mention); relation to S35, S52.

Main results proved. Coherence of the étale topos of a qcqs scheme.

Role in the course. *Depends on* S35, S52, S58. *Feeds* S60–S61. *Arc:* closes the loop: the étale world has enough points.

Unifying thread. *Geometric points suffice to see everything étale — Deligne in geometry.*

References. Mac Lane–Moerdijk ch.VII; SGA4 exp.VI; Milne ch.II.

11 Postlude — Synthesis and the Big Picture

11.1 Session 60 — The Point: One Concept, Many Worlds

Content to cover.

- Recap of all incarnations: prime, valuation, prime ideal, closed/generic, T -valued point, place, geometric morphism, model, non-commutative glimpse.
- The big summary table (below); Dessins d’Enfants: $G_{\mathbb{Q}}$ acting on children’s drawings (via S35).
- Final message: the point is a precise structure — evaluation from a local perspective.

World	“Point” is	Evaluation is
Number theory	prime / place	residue, $ \cdot _p$
Geometry	prime ideal / closed-generic	$\kappa(\mathfrak{p})$
Topology	point of a space	stalk
Topos theory	geom. morphism $\mathbf{Set} \rightarrow \mathcal{E}$	p^*
Logic	model of a theory	truth
Non-comm.	character / state	expectation

Role in the course. *Depends on* everything. *Feeds* S61. *Arc:* the payoff: one structure, many disguises.

References. Grothendieck, *Esquisse*; Schneps.

11.2 Session 61 — Points in Mechanics, Quantum Physics, and Langlands

Content to cover.

- Points as states in classical mechanics (phase space); states as “points” of non-commutative algebras in QM (echo of S51).
- The Langlands program as non-abelian reciprocity (template S47); p -adic Hodge theory; geometric Langlands & ∞ -topoi; perfectoid spaces; arithmetic topology (knots $\leftrightarrow$ primes).
- Open problems & reading paths per participant.

Role in the course. *Depends on* S60. *Feeds* future study. *Arc:* the research map; audience-tailored horizon.

References. Connes; Frenkel; Scholze; Mazur.

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End of Detailed Syllabus — Point: From Number Theory to Algebraic Geometry and Logic
Version 2.1 (Merged Maximal Edition) — 64 meetings (P1–P3 + S1–S61)